LAB SHEET - 9

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Section: C
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```
!pip install ultralytics
```

```
Collecting ultralytics
  Downloading ultralytics-8.4.174-py3-none-any.whl.metadata (46 kB)
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Downloading ultralytics_thop-2.2.2-py3-none-any.whl (32 kB)
Installing collected packages: nvidia-ml-py, ultralytics-thop, ultralytics-platform, ultralytics
Successfully installed nvidia-ml-py-13.615.71 ultralytics-8.4.174 ultralytics-platform-0.1.82 ultralytics-thop-2.2.2
```

```
from ultralytics import YOLO
import cv2
import matplotlib.pyplot as plt

print("Libraries imported successfully")
```

```
Libraries imported successfully
```

```
image_path = "/content/Golden Retriever.png"

print("Image path:", image_path)
```

```
Image path: /content/Golden Retriever.png
```

```
model = YOLO("yolov8n.pt")

print("YOLOv8 model loaded successfully")
```

```
YOLOv8 model loaded successfully
```

```
img = cv2.imread(image_path)

if img is None:
    print("Error: Image could not be loaded")
else:
    print("Image loaded successfully")

    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    plt.figure(figsize=(10, 7))
    plt.imshow(img_rgb)
    plt.axis("off")
    plt.title("Original Image")
    plt.show()
```

```
Image loaded successfully
```




```
results = model(image_path)

print("Object detection completed successfully")
```

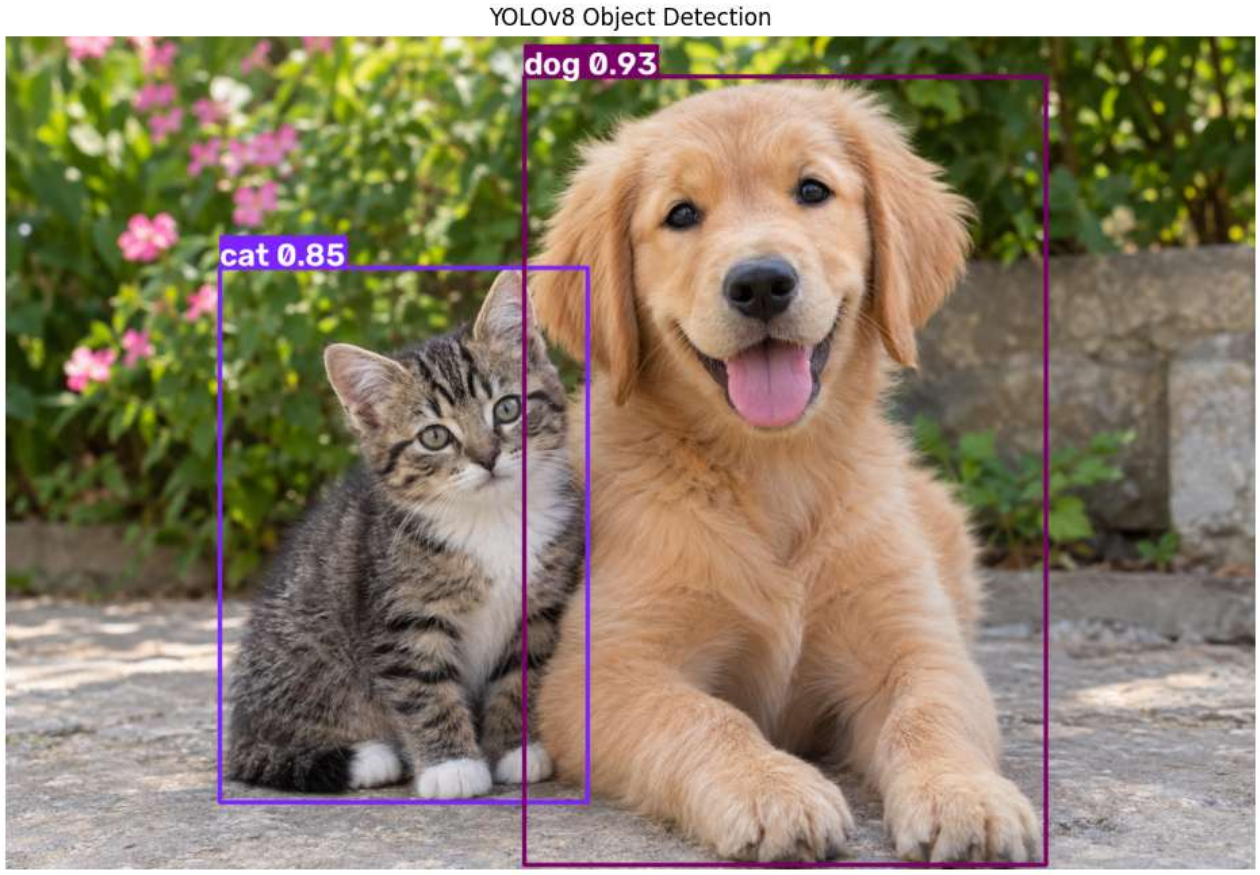
```
image 1/1 /content/Golden Retriever.png: 448x640 1 cat, 1 dog, 184.8ms
Speed: 4.6ms preprocess, 184.8ms inference, 33.3ms postprocess per image at shape (1, 3, 448, 640)
Object detection completed successfully
```

```
result_image = results[0].plot()

result_image_rgb = cv2.cvtColor(
    result_image,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(12, 8))
```

```
plt.imshow(result_image_rgb)
plt.axis("off")
plt.title("YOLOv8 Object Detection")
plt.show()
```




```
print("Detected Objects")
print("=" * 40)

if len(results[0].boxes) == 0:
    print("No objects detected")
else:
    for box in results[0].boxes:
        class_id = int(box.cls[0])
        confidence = float(box.conf[0])
        class_name = model.names[class_id]

        print(
            f"Object: {class_name} | "
            f"Confidence: {confidence:.2f}"
        )
```

```
Detected Objects
========================================
Object: dog | Confidence: 0.93
Object: cat | Confidence: 0.85
```

```
object_counts = {}

for box in results[0].boxes:
    class_id = int(box.cls[0])
    class_name = model.names[class_id]

    object_counts[class_name] = object_counts.get(class_name, 0) + 1

print("Object Count")
print("=" * 40)

if len(object_counts) == 0:
    print("No objects detected")
```

```
else:
    for name, count in object_counts.items():
        print(f"{name}: {count}")
```

```
Object Count
========================================
dog: 1
cat: 1
```

```
print("Bounding Box Coordinates")
print("=" * 50)

if len(results[0].boxes) == 0:
    print("No bounding boxes detected")
else:
    for box in results[0].boxes:
        class_id = int(box.cls[0])
        class_name = model.names[class_id]
        confidence = float(box.conf[0])

        x1, y1, x2, y2 = box.xyxy[0].tolist()

        print(
            f"{class_name} | "
            f"Confidence: {confidence:.2f} | "
            f"Coordinates: "
            f"({x1:.0f}, {y1:.0f}, {x2:.0f}, {y2:.0f})"
        )
```

```
Bounding Box Coordinates
==================================================
dog | Confidence: 0.93 | Coordinates: (637, 49, 1279, 1019)
cat | Confidence: 0.85 | Coordinates: (263, 284, 716, 941)
```

```
if len(results[0].boxes) > 0:
    confidences = results[0].boxes.conf.cpu().numpy()
    average_confidence = float(confidences.mean())

    print(
        "Average Detection Confidence:",
        round(average_confidence, 2)
    )
else:
    average_confidence = 0.0
    print("No objects detected")
```

```
Average Detection Confidence: 0.89
```

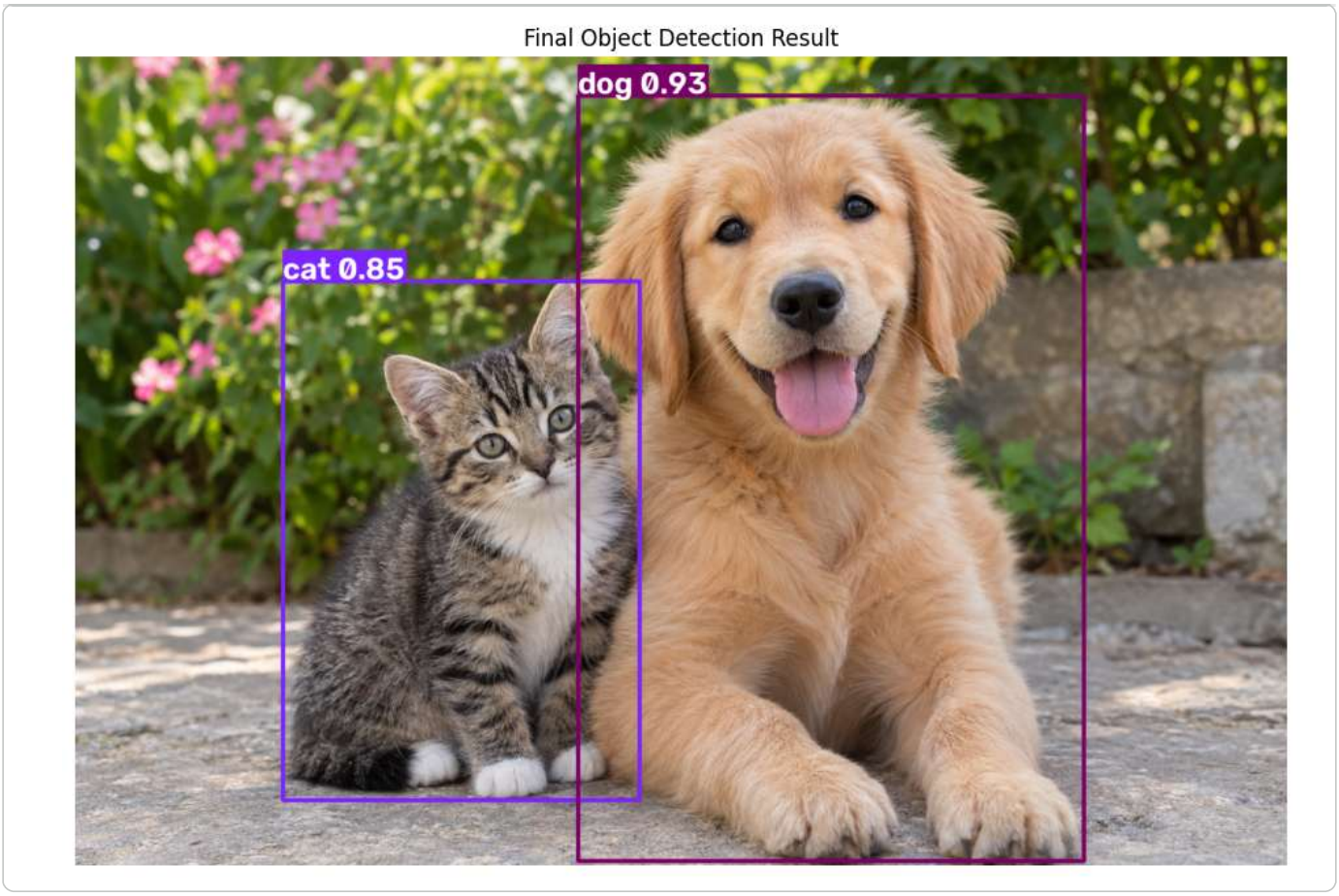
```
output_path = "/content/object_detection_result.jpg"

cv2.imwrite(
    output_path,
    result_image
)

print("Detection result saved successfully")
print("Output path:", output_path)
```

```
Detection result saved successfully
Output path: /content/object_detection_result.jpg
```

```
plt.figure(figsize=(12, 8))
plt.imshow(result_image_rgb)
plt.axis("off")
plt.title("Final Object Detection Result")
plt.show()
```



Summary

Model Used: YOLOv8n
Dataset: MS COCO
Input Image: Dog and Cat image
Number of Objects Detected: 2
Detected Objects: Dog, Cat
Dog Detection Confidence: 0.93
Cat Detection Confidence: 0.85
Average Detection Confidence: 0.89

Result: YOLOv8 successfully detected and recognized the dog and cat in the input image using bounding boxes and confidence scores.